

ANUJ WAGHULDE

Software Engineer | ICPC Regionalist 2023–24 | Networking, Backend, Embedded Systems & CI/CD

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SUMMARY

Software Engineer with 3+ years designing and architecting backend, embedded, and networked systems for LTE/5G router platforms. Proven track record building reusable Python/C++ frameworks, REST APIs, and system integrations; architecting end-to-end CI/CD pipelines and release automation; and developing AI-agent tooling (Claude Code, MCP, RAG) that automates CVE triage, network diagnostics, and CI/CD failure investigation.

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, OOP, DSA, Multithreading, Multiprocessing, Async Programming

Backend & APIs: REST APIs, API Design, Flask, Microservices, Service Integration

Databases: PostgreSQL, MySQL, MongoDB, SQL, Firebase

AI & Agentic Engineering: LLMs, AI Agents, MCP, RAG, Tool Orchestration, Prompt Engineering, Claude Code, Cursor

Networking & Embedded: Computer Networks, TCP/IP, DHCP/DNS, DHCP Relay, DDNS, IPsec VPN, Firewall, BGP, QoS, Certificates, LTE/5G, Cellular Modules, Callbox, WAN/LAN, SSH/Telnet, GNSS, IPv4/IPv6

CI/CD & DevOps: Git, GitHub Actions, Jenkins, GitLab CI, Docker, Grafana, Plotly, Pytest, Robot Framework, Selenium, Linux

EXPERIENCE

Software Engineer

July 2024 – Present

Semtech Advanced Systems

- **Architect reusable Python/C++ software components and frameworks** for embedded LTE/5G router platforms, integrating REST APIs, device configuration, firmware operations, diagnostics, monitoring, and system services.
- **Develop and integrate features across AirLink embedded router platforms** and cellular modules — device management, cloud connectivity, admin/status interfaces, event reporting, firmware upgrade/downgrade, and remote access.
- **Develop cellular integration and callbox workflows** for LTE/5G radio/profile changes, module state, network registration/attach, IP connectivity, and repeatable feature execution across integrated systems.
- **Architect end-to-end GitHub Actions CI/CD pipelines** for environment provisioning, firmware deployment, artifact/report publishing, and release workflows, reducing release-cycle time by **35%**.
- **Engineer concurrent and asynchronous execution** using multiprocessing, TCP/session coordination, and shared-resource locking to coordinate callbox, Wi-Fi, VPN, and other shared infrastructure across parallel workflows.
- **Build and apply AI-agent tooling** with Claude Code, MCP, and RAG to engineering automation — including a tool-driven diagnostic agent for network and device investigation — spanning CVE analysis, network diagnostics, and CI/CD failure investigation across tools, code, logs, and device data.
- **Implement, integrate, and debug networking features** including DHCP/DNS, DHCP Relay, DDNS, BGP, IPsec VPN, Firewall, QoS, and certificates, tracing issues across APIs, device state, configuration, logs, and network behavior; drove resolution of **25+ critical defects** before release.
- **Increase automated execution coverage from 60% to 80%** by extending reusable components, validation flows, diagnostics, and failure handling; added supporting Grafana/Plotly observability for pipeline and system insights.

Software Engineer Intern

July 2023 – June 2024

Semtech Advanced Systems

- **Developed Python automation scripts** for firmware upgrade/downgrade, device configuration, connectivity checks, log collection, and basic feature validation on LTE/5G router platforms.
- **Automated simple device and networking feature workflows**, improving repeatability of routine validation tasks and regression checks.
- **Built Flask/Plotly dashboards and utilities** to visualize automation results, device status, and execution data.

PROJECTS

AI-Powered CVE Analysis Pipeline — Headless Security Automation

Claude Code, Python, LLMs, BlackDuck, NVD, OSV, Yocto

- **Built a headless CI security workflow** to analyze CVEs with CVSS ≥ 7.0 across embedded Linux platforms, correlating BlackDuck findings with NVD, OSV.dev, and Yocto/BitBake metadata.
- **Implemented a staged flow** for CVE retrieval, source inspection, patch generation, validation, reporting, and PR creation; handles conflicts through NEEDS_REVIEW and avoids automatic merging.
- **Added evidence-backed NOT_AFFECTED decisions** recording searched files, analysis method, supporting evidence, and conclusion for auditable security triage.

NetworkOps Copilot — LLM-Based Network Diagnostics Agent

Claude Code, Python, LLMs, MCP, RAG, REST, SSH

- **Designed a tool-driven diagnostic agent** where an engineer provides a device hostname/IP and platform; controlled REST/SSH and diagnostic tools inspect reachability, interfaces/routes, VPN, services, configuration, DHCP/DNS, and logs.
- **Added pre-flight checks and RAG** over runbooks, device documentation, configuration references, and known failure patterns to validate prerequisites and correlate live device evidence with retrieved context.
- **Supports scheduled monitoring** via a separate cron-based job after validation, reporting later failures with the failed check, supporting evidence, and likely cause; operational changes remain behind explicit human approval.

Release Intelligence — AI-Assisted CI/CD Failure Analysis

Claude Code, Python, GitHub Actions, LLMs, MCP, Grafana

- **Built an AI-assisted investigation layer** over CI results, logs, artifacts, and Grafana dashboards; retrieves workflow metadata, device logs, dumps, screenshots, metrics, and execution context for a selected run or failure.
- **Correlates failures across workflow stages, build metadata, device state, Git diffs/PRs, and historical runs** to identify relevant fault domains across build, deployment, device, network, environment, and code changes.
- **Uses MCP/LLM-driven investigation** to trace failures into relevant code and pull requests and produce structured summaries with evidence, affected components, related changes, and next investigation paths.

EDUCATION

Bachelor of Engineering (B.E.), Computer Engineering - Data Science (Hons)

2020 - 2024

PCET's Pimpri Chinchwad College of Engineering and Research, Ravet, Pune

CGPA: 9.2/10

ACHIEVEMENTS

- **ICPC Regionalist 2023-24** — Qualified for the Kanpur and Amritapuri Regional Contests; 4-Star CodeChef, Knight LeetCode (Top 3%), and Meta Hacker Cup Round 1.
- **TIAA Hackathon Winner** — Won a competitive hackathon for full-stack application development and solution design.
- **Competitive Achievements** — GATE 2024 Qualified; 5th Rank in MaTPO Programming Idol.
- **Patent Granted** — “Language Learning Robotics Assistant”, Application No. 415490-001.
- **Research & Publications** — Published research on Android healthcare and mental-health detection/recommendation; copyright registered.